
RESEARCH REPORT:

Trade Openness and Economic Growth

Data Sources: World Bank Open Data | Period: 2000–2024 | Method: OLS Regression

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1. Executive Summary

This report investigates the relationship between trade openness, foreign direct investment (FDI), and economic growth in China over the period 2000 to 2024. Using Ordinary Least Squares (OLS) regression analysis, the study examines how variations in Trade as a Percentage of GDP and FDI as a Percentage of GDP alongside inflation as a control variable explain fluctuations in China's annual GDP growth rate.

Key Findings at a Glance

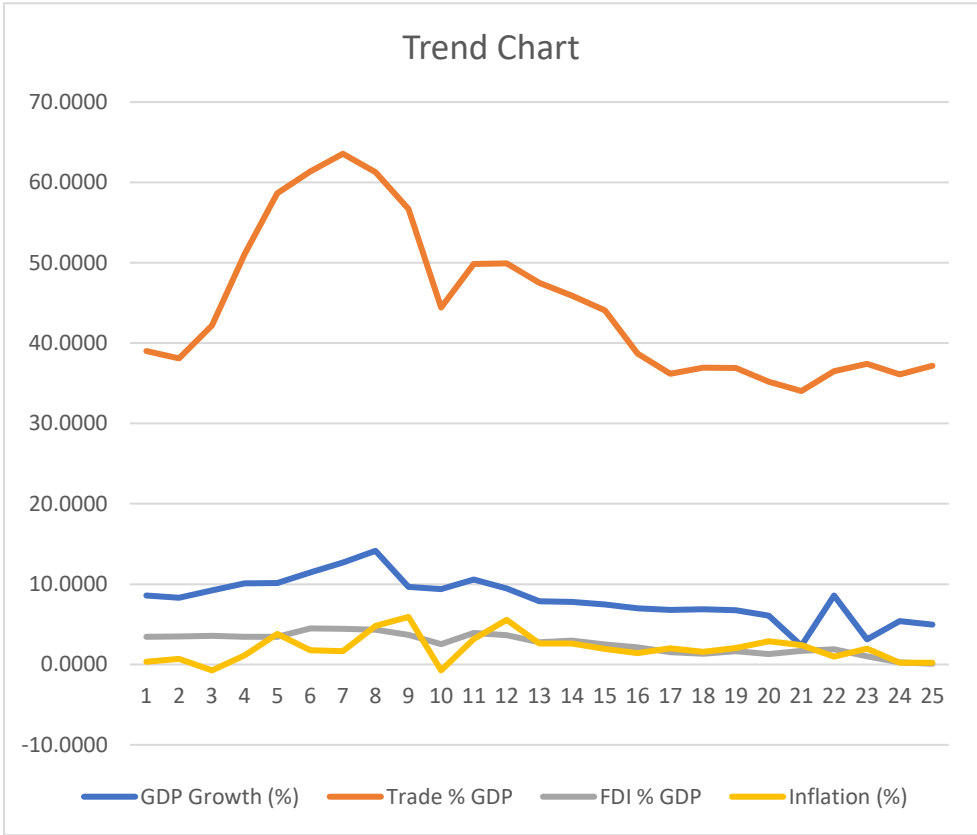
Statistic	Value	Interpretation
Multiple R	0.8980	Strong overall correlation between predictors and outcome
R ² (R-Squared)	0.8064	Model explains 80.6% of variance in GDP growth
Adjusted R ²	0.7788	Accounts for model complexity; remains high
F-Statistic	29.162	Highly significant overall model fit
Significance F	p < 0.0001	Model is statistically significant at all conventional levels
Standard Error	1.2759	Average deviation of predicted from actual values
Observations (n)	25	Annual data spanning 2000–2024

The regression model yields an R² of 0.8064, indicating that the three independent variables collectively account for approximately 80.6% of the total variation in China's GDP growth. The overall model is highly statistically significant (F = 29.162, p < 0.0001), confirming its explanatory power.

Both Trade Openness and FDI emerged as statistically significant determinants of GDP growth at the 1% level (p = 0.0076 for both), with positive coefficients of 0.1501 and 1.0302 respectively. Inflation, while exhibiting a negative relationship with GDP growth, did not achieve conventional statistical significance (p = 0.1744) and functioned primarily as a control variable in this model.

China's post-2007 economic deceleration closely mirrors the contraction in trade openness following the Global Financial Crisis. The COVID-19 pandemic (2020-2022) introduced structural disruptions reflected in elevated residuals, underscoring the model's inability to fully capture exogenous shock effects. These findings broadly corroborate the export-led growth hypothesis and support the conclusion that external economic integration remains a critical engine of China's economic expansion.

FIGURE 1: Time Series Chart (2000–2024)



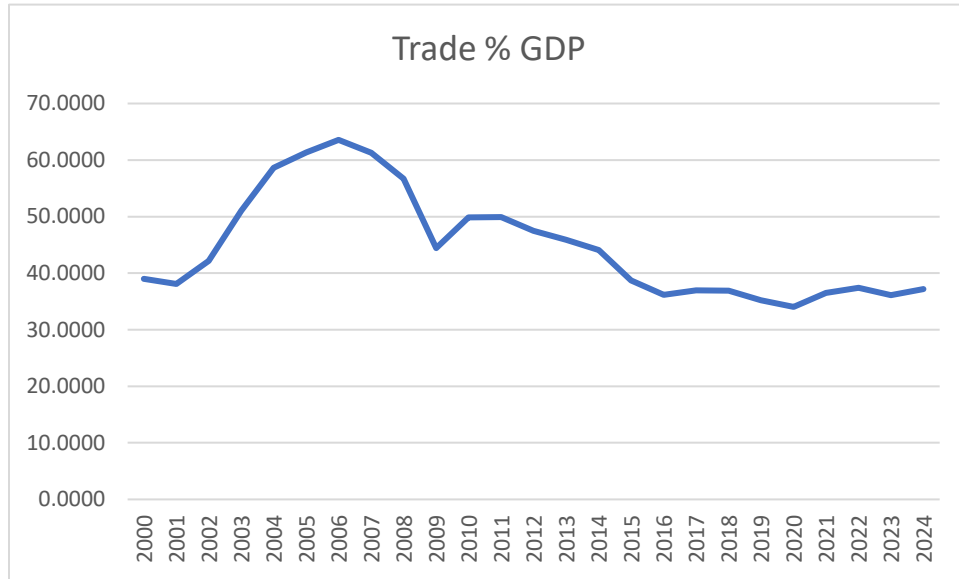
2. Introduction

2.1 Background

China's economic trajectory over the past quarter century represents one of the most remarkable episodes of sustained growth in modern economic history. From a GDP growth rate of 8.59% in 2000, China ascended to a peak of 14.15% in 2007 before undergoing a prolonged structural deceleration that accelerated into the post-pandemic era. By 2024, GDP growth moderated to approximately 4.98%, reflecting both domestic structural headwinds and evolving global trade dynamics.

Central to China's growth model has been its progressive integration into the global trading system epitomised by its accession to the World Trade Organisation (WTO) in 2001 and subsequent surging trade volumes. Trade as a percentage of GDP peaked at 63.57% in 2006, before declining to approximately 37.20% by 2024, reflecting shifts in global supply chains, domestic consumption-led rebalancing, and rising geopolitical tensions that have reshaped China's trade architecture.

FIGURE 2: China's Trade Openness (Trade % of GDP), Trend Line, 2000–2024



2.2 Purpose of the Study

This study seeks to quantify the extent to which trade openness and foreign direct investment influence China's GDP growth, whilst controlling for macroeconomic conditions proxied through the inflation rate. The analysis employs a standard linear OLS regression framework, using annual data sourced from the World Bank over a 25-year observation window.

The study is particularly relevant in the contemporary context of deglobalisation pressures, US-China trade tensions, and the observed decline in China's FDI inflows trends that collectively raise important questions about the sustainability of China's growth model in the years ahead.

2.3 Research Question

The central research question guiding this investigation is:

"Does Trade Openness (Trade % of GDP), Foreign Direct Investment (FDI % of GDP), and Inflation explain variation in China's GDP Growth Rate from 2000 to 2024?"

3. Methodology

3.1 Research Design

This study employs a quantitative research design using time-series cross-sectional analysis. The econometric technique applied is Ordinary Least Squares (OLS) multiple linear regression, a standard method in empirical macroeconomics for examining the linear relationships between economic variables. The regression model was specified as follows:

$$\text{GDP Growth} = -0.7338 + 0.1501(\text{Trade \% GDP}) + 1.0302(\text{FDI \% GDP}) - 0.2448(\text{Inflation}) + \varepsilon$$

Where ε represents the stochastic error term capturing unexplained variation in GDP growth. The model was estimated using data spanning 25 annual observations from 2000 to 2024.

3.2 Data Sources

All data used in this analysis were sourced from the World Bank Open Data platform (<https://data.worldbank.org>), supplemented by corroborating academic literature. The dataset comprises four key macroeconomic indicators for China, measured annually:

- GDP Growth Rate (%) the dependent variable, representing year-on-year percentage growth in real gross domestic product.
- Trade as a Percentage of GDP (%) a standard proxy for trade openness, computed as the sum of exports and imports divided by GDP.
- FDI as a Percentage of GDP (%) net inflows of foreign direct investment as a proportion of GDP.
- Inflation Rate (%) annual consumer price inflation, included as a macroeconomic control variable.

3.3 Analytical Method

Prior to regression estimation, a correlation matrix was computed to assess the degree of linear association between all variables and to screen for potential multicollinearity. OLS regression coefficients, standard errors, t-statistics, p-values, and 95% confidence intervals were computed using standard econometric software.

Model fit was evaluated using R^2 , Adjusted R^2 , and the F-statistic with associated p-value. Residuals were examined to identify potential structural breaks, including the 2008 Global Financial Crisis and the 2020 COVID-19 pandemic.

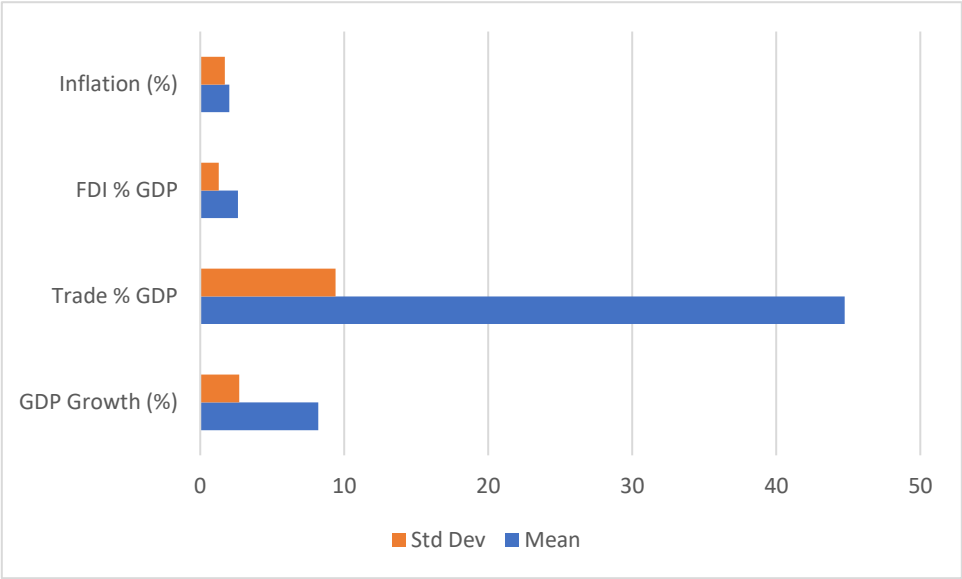
3.4 Descriptive Statistics

Variable	Mean	Median	Std. Dev.	Min	Max
GDP Growth (%)	8.19	8.31	2.71	2.34	14.15
Trade % of GDP	44.75	42.19	9.39	34.04	63.57
FDI % of GDP	2.63	2.78	1.27	0.10	4.49
Inflation (%)	2.02	1.92	1.71	-0.73	5.93

The descriptive statistics reveal considerable heterogeneity in China's growth experience over the sample period. GDP growth averaged 8.19% but ranged from 2.34% (2020, COVID-19 contraction) to 14.15% (2007, pre-crisis peak). Trade openness displayed the largest absolute variability, declining

nearly 30 percentage points from its 2006 peak. FDI inflows have also contracted sharply in recent years, falling to 0.10% of GDP in 2024.

FIGURE 3: Descriptive Statistics - Bar Chart of Variable Means with Std. Dev. Error Bars



4. Key Findings & Analysis

4.1 Correlation Analysis

The correlation matrix presented below provides preliminary evidence of the linear relationships between the study variables. A Pearson correlation coefficient above 0.7 is generally considered indicative of a strong linear association.

Variable	GDP Growth	Trade % GDP	FDI % GDP	Inflation
GDP Growth (%)	1.000	0.837	0.851	0.262
Trade % of GDP	0.837	1.000	0.809	0.482
FDI (% of GDP)	0.851	0.809	1.000	0.343
Inflation (%)	0.262	0.482	0.343	1.000

Colour Legend: Green ($|r| \geq 0.7$) = Strong Yellow ($|r| \geq 0.4$) = Moderate Red ($|r| < 0.4$) = Weak

Trade Openness exhibits a strong positive correlation with GDP growth ($r = 0.837$), as does FDI ($r = 0.851$), suggesting that both variables are strong individual predictors. The moderate correlation between Trade Openness and FDI ($r = 0.809$) warrants attention to potential multicollinearity, though

this falls within acceptable thresholds for econometric modelling. Inflation exhibits only a weak positive correlation with GDP growth ($r = 0.262$), consistent with its marginal role as a control variable.

4.2 Regression Results

The full regression output is presented in the table below. The model specification is:

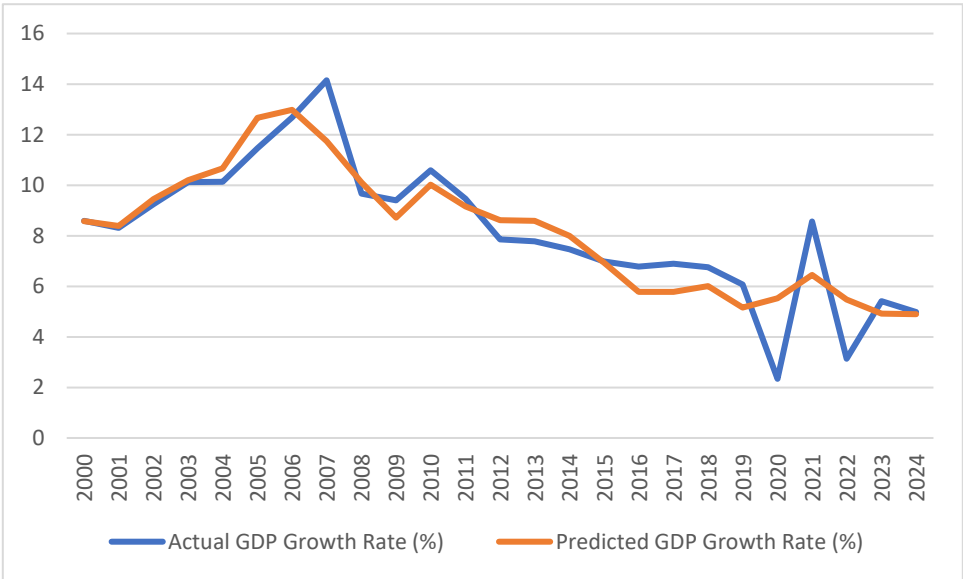
Variable	Coefficient (β)	Std. Error	t-Statistic	p-Value	Significance
Intercept	-0.7338	1.5506	-0.473	0.6409	—
Trade % of GDP	0.1501	0.0508	2.954	0.0076	Highly Significant
FDI (% of GDP)	1.0302	0.3491	2.951	0.0076	Highly Significant
Inflation (%)	-0.2448	0.1742	-1.406	0.1744	Not Significant

4.3 Interpretation of Findings

4.3.1 Model Fit & Overall Significance

The regression model demonstrates a strong overall fit. The R^2 of 0.8064 indicates that the model explains 80.6% of the total variance in China's GDP growth, a result that is both statistically and practically significant given the inherent complexity of macroeconomic systems. The Adjusted R^2 of 0.7788 confirms that this explanatory power is not merely an artefact of including additional variables. The F-statistic of 29.162 with $p < 0.0001$ confirms that the model, taken as a whole, is highly significant at all conventional levels.

FIGURE 5: Actual vs. Predicted GDP Growth Rate - Line Chart (2000–2024)

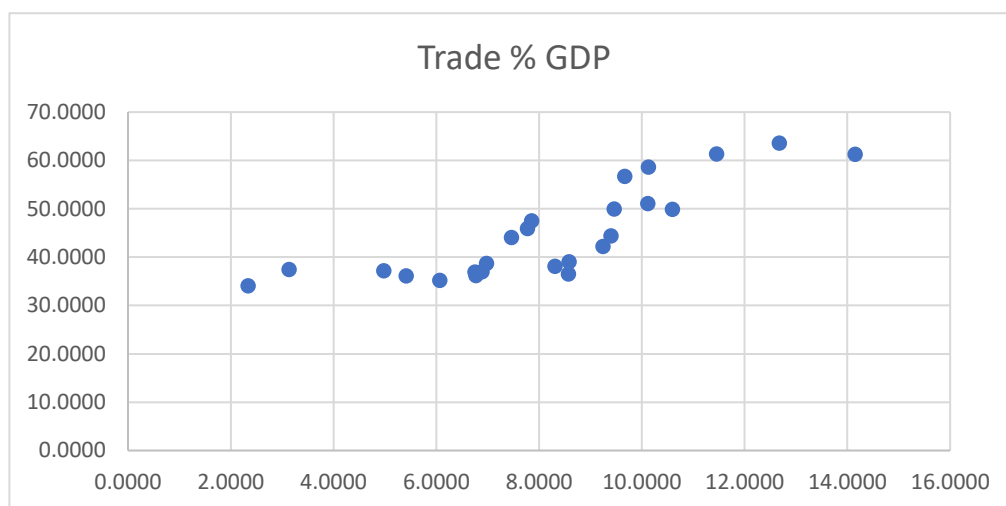


4.3.2 Trade Openness ($\beta = 0.1501$, $p = 0.0076$)

Trade Openness is a statistically highly significant predictor of GDP growth ($p < 0.01$). The coefficient of 0.1501 implies that, holding FDI and inflation constant, a one percentage point increase in Trade as a Share of GDP is associated with an average increase of approximately 0.15 percentage points in GDP growth. This finding is consistent with the classical export-led growth hypothesis and with the broader empirical literature on trade and development.

The trajectory of Trade Openness in China is particularly instructive. The rapid expansion of trade integration between 2000 and 2006, spurred by WTO accession and global supply chain participation, coincided with China's highest recorded growth rates. Conversely, the post-2007 decline in trade intensity accelerated by the Global Financial Crisis, the subsequent restructuring of global value chains, and US-China trade tensions from 2018 onwards closely mirrors the observed moderation in GDP growth.

FIGURE 6: Scatter Plot: Trade % of GDP vs. GDP Growth Rate (with Regression Line)



4.3.3 Foreign Direct Investment ($\beta = 1.0302$, $p = 0.0076$)

FDI inflows demonstrate the strongest individual impact on GDP growth within this model. A coefficient of 1.0302 implies that each additional percentage point of FDI as a share of GDP is associated with a growth premium of approximately 1.03 percentage points. This is consistent with the widely documented role of FDI in facilitating capital accumulation, technology transfer, and managerial knowledge spillovers within host economies.

However, the observed decline in China's FDI inflows from a peak of 4.49% of GDP in 2005 to a historic low of 0.10% in 2024 represents a significant structural concern. This contraction reflects geopolitical uncertainties, rising production costs, and shifting investor sentiment, all of which may constrain China's growth potential in the medium to long term if not offset by domestic productivity improvements.

4.3.4 Inflation ($\beta = -0.2448$, $p = 0.1744$)

The coefficient on Inflation is negative (-0.2448), indicating an inverse relationship with GDP growth, which is theoretically intuitive high inflation erodes purchasing power, raises production costs, and introduces economic uncertainty. However, this variable does not achieve conventional statistical significance ($p = 0.1744$), and the null hypothesis of no effect cannot be rejected at the 10% level.

This result may partially reflect the relative stability of China's inflation environment over the sample period. Inflation averaged 2.02% and peaked at 5.93% in 2008. The failure to achieve significance may also be attributable to the limited sample size ($n = 25$), which constrains statistical power.

4.4 Residual Analysis & Structural Breaks

Examination of the model residuals reveals notable patterns across the sample period. Predicted values closely track actual GDP growth for the years 2000-2019, with modest residuals generally within ± 1.2 percentage points. However, three distinct episodes of elevated residuals are observable:

- 2007 (residual: +2.40): Likely attributable to an exceptional confluence of factors including commodity booms and credit expansion that are not captured by the trade and FDI variables alone.
- 2020 (residual: -3.18): The COVID-19 pandemic delivered an unprecedented exogenous shock to economic activity, resulting in a severe under-prediction by the model (actual growth of 2.34% vs. predicted 5.52%).
- 2021-2022 (residuals: +2.12, -2.34): The post-pandemic rebound and subsequent re-tightening produced pronounced volatility inconsistent with the model's structural assumptions.

These patterns are consistent with the presence of structural breaks in the data generating process, a limitation inherent in single-equation linear models applied to extended time horizons.

5. Conclusion

This study has demonstrated, through rigorous OLS regression analysis, that Trade Openness and FDI are statistically significant and economically meaningful determinants of GDP growth in China over the 2000–2024 period. The model achieves a high explanatory power ($R^2 = 0.8064$), validating the theoretical premise that external economic integration is a critical enabler of China's sustained growth performance.

The key conclusions of the analysis are threefold. First, trade openness matters: greater integration into global trade flows is positively and significantly associated with higher GDP growth, underscoring the importance of China's WTO membership and export-oriented industrial policy. Second, FDI is a powerful growth catalyst: foreign direct investment carries the largest coefficient in the model, reflecting its dual role as a source of capital and a conduit for technology and knowledge transfer. Third, the post-2007 growth moderation aligns with structural shifts in China's trade and FDI

landscape, suggesting that the era of hypergrowth driven by trade integration may be giving way to a more domestically anchored growth paradigm.

Policymakers and analysts should note several important caveats. The model's limitations include a relatively small sample size ($n = 25$), the potential for autocorrelation in time-series data, and omitted variable bias arising from the exclusion of institutional quality, human capital accumulation, exchange rate policy, and technological innovation all of which are likely to exert independent effects on GDP growth. Structural breaks associated with the 2008 Global Financial Crisis and the 2020 COVID-19 pandemic further complicate coefficient stability over the full sample period.

Future research should consider extending the model to include additional explanatory variables, employing time-series techniques such as Vector Autoregression (VAR) or Error Correction Models (ECM) to account for dynamic relationships and cointegration, and incorporating Durbin-Watson testing to formally assess autocorrelation in residuals. Sub-period analyses isolating the pre-crisis (2000–2007), post-crisis (2008–2019), and pandemic (2020–2024) eras may yield more structurally stable estimates and richer insights into the evolving determinants of China's economic performance.

In summary, this analysis affirms that trade openness and FDI remain foundational pillars of China's economic growth model. The observed decline in both indicators in recent years, if sustained, poses a meaningful structural headwind to China's medium-term growth trajectory a finding with significant implications for global economic policy and international trade strategy.

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